

Review

An Evolutionary Game Analysis of the Relationship between Core Stakeholders of Forest Governance

Huiyong Xu ¹, Xuejiao Zhao ² and Dahong Zhang ^{1,*}

¹ College of Economics and Management, Beijing Forestry University, Beijing 100083, China; xuhy1988@126.com

² Research Center for Green Development of Great Wall Cultural Economic Belt, School of Tourism, Collaborative Innovation Center for Urban-Rural Integration Development, New Rural Development Center, Hebei University of Economics and Business, Shijiazhuang 050061, China; zhxj19@hueb.edu.cn

* Correspondence: zhangdahong591120@163.com

Abstract: The interest relationship among forest governance stakeholders will change with the influence of political, economic, cultural, and other factors. Studying the logical relationship between various stakeholders in the process of forest management is an important measure to adhere to the “clear water and green mountains are gold hills and silver mountains” and implement the concept of green development under the new form. In this paper, the evolutionary game method is used to build a dynamic evolutionary game model, analyze the evolutionary stability strategy and replicator dynamics of the three core stakeholders of government, enterprises, and rural community residents, and calculate the equilibrium stability point. The behavioral strategy selection and evolutionary stability conditions of each game player are analyzed. The results show that the “guidance” of the government is a strong guarantee for the realization of the tripartite equilibrium. The government is in a very active and strong position and needs the guidance of the government to achieve stability. The results show that the “guidance” of the government is a strong guarantee for the realization of the tripartite equilibrium. The government is in a very active and strong position and needs the guidance of the government to achieve stability. Therefore, the government should actively encourage rural community residents to participate in forest management, strengthen the supervision of forestry enterprises, and achieve effective management of forest resources.



Citation: Xu, H.; Zhao, X.; Zhang, D. An Evolutionary Game Analysis of the Relationship between Core Stakeholders of Forest Governance. *Sustainability* **2023**, *15*, 10546. <https://doi.org/10.3390/su151310546>

Academic Editor: Luigi Aldieri

Received: 25 May 2023

Revised: 28 June 2023

Accepted: 3 July 2023

Published: 4 July 2023



Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: forest governance; government guidance; core stakeholders; evolutionary game

1. Introduction

The forestry governance system is an important component of the national governance system, and forest governance is an important part of the forestry governance system. During the 13th Five-Year Plan period, China’s forest coverage rate increased to 23.04%, and forest stock volume exceeded 17.5 billion cubic meters, maintaining “double growth” for 30 consecutive years, becoming the country with the largest increase in forest resources (Department of Ecology of the State Forestry and Grassland Administration: China’s forest stock volume has exceeded 17.5 billion cubic meters for 30 consecutive years to maintain a growth trend. <http://www.forestry.gov.cn/zlszz/4253/20210127/102959290529535.html>, accessed on 1 May 2022). However, the problems of single tree species and low stand quality still need to be solved by expanding the increment scale, optimizing the stock structure, and other forest management methods, so as to promote the modernization level of forestry management. In order to better manage forests, it is clear that government policies alone, or even market and technical measures, are not enough. This involves the concept of governance. The research on governance reform is mainly concentrated in the fields of institutional analysis and public policy, but it is still a new thing to introduce governance principles into forest management and put forward forest governance. The efficiency principle pursued by forest management means that the best management level

of a forest should be achieved no matter what kind of process it goes through. The public space involved in forest governance in the environmental era not only refers to the unilateral “management” or “governance” of the forest by the government but also emphasizes that the values of each relationship subject can interweave, collide, and run in with each other, finally seeking a compromise relationship or an appropriate governance approach. At the same time, the government and the market provide a platform to coordinate the relationship between various subjects in another capacity, so as to reach dialogue, cooperation, or mutual understanding (Zhang Yujun, Tourism Research Center, Chinese Academy of Social Sciences, says ecotourism can play an important role in forest governance systems. https://mp.weixin.qq.com/s?src=11×tamp=1613314523&ver=2890&signature=3QOobHdTG9ftIHxArOrLWOXjVNAEDV1Wl0FklMuW93YVkcIVMRnEg4-smSOzTaE4mVMUHwqtRkKOeaSKtlp5qjNcKEdiFevSGehkPUlti13qgMsxtY*xd-UtF7vK2i8&new=1, accessed on 1 May 2022). In the process of forest governance, various stakeholders have different interests and demands, and conflicts and contradictions are increasingly prominent in the course of development. Hengst-Ehrhart Y and Schraml U [1] argue that most of the research content introduced by stakeholder theory into the forestry field focuses on the conflicts among stakeholders and the factors that affect the actions of stakeholders in forestry management. Salam, MA, Noguchi [2] conducted research on stakeholders’ ability to manage forest resources; Schusser, Krott, Movuh, et al. [3] conducted research on stakeholders for the development of community forestry; Kumar S and Kant S [4] conducted research on the forest value preference of forestry stakeholders. Therefore, it is urgent to increase theoretical and empirical research on various stakeholders in forest management. Through analysis, we can understand the interest demands of all stakeholders, timely solve the interest conflicts of all stakeholders in forest governance, coordinate the relationship between all stakeholders, and promote the sustainable development of forest resources.

2. Literature Review

In order to have a more comprehensive understanding of foreign scholars’ research on forest governance, this paper uses the ScienceDirect of the Elsevier Archive as the engine and “forest governance” as the search term. By 2023, there are a total of 840 literatures. This fully shows that foreign countries are relatively mature in forest management. In terms of forest governance, the research objects include urban forest governance, community forest management, global forest governance, NGO participation, decentralization of forest governance, and many other aspects. In terms of research content, forests become an important stage for the game between different actors, and the power relationship between actors profoundly affects the effect of forest governance. Most studies of forest governance have led to decentralization, giving more power to the grass roots. Ahmad Abedi Sarvestani [5] argues that there is a preference for closed co-management arrangements, whereby government and non-government actors manage forests together. Arts, Bas [6], in discussing the various implications and major criticisms of forest governance, argue that a key assumption in many management studies is that the state has largely withdrawn from the forest sector and that forest governance has shifted from the state to the market and society. Juerges Nataly [7] argues that the transfer of power from government actors to civil society and market actors in forest governance research has been the subject of intense debate in the scientific community. By analyzing how ongoing transitions and power shifts in forest governance affect power relations among actors interested in various ecosystem services (ESS) in nine countries (Germany, Ireland, Italy, Lithuania, Portugal, Slovakia, Sweden, the Netherlands, and Turkey), it is concluded that government actors are the most powerful actors in most countries, Hence the promotion of forest management. Agbo Victor Mawutor and Swatuk Larry [8] sampled forest stakeholders in Kwabeng, the administrative capital of the Atewa West District, to understand forest governance challenges and outline strategies for overcoming them. The study revealed that a bottom-up, all-inclusive approach to managing forest resources is necessary. Ewane Basil Ewane [9] considers Effective implementation of forest governance principles such as

transparency, participation, accountability, coordination, capacity, user rights protection, and security driven by positive government interventions is instrumental for sustainable forest management. Salo Matti [10] focuses on how different prevailing decision-making and administrative rationalities, or governmentalities, have shaped the knowledge base and epistemic requirements of forest governance in Peru. Cadman Timothy [11] outlines the development of community forestry, the legislative, regulatory, and governance frameworks underpinning this unique system of community-driven forest management, and its relationship to sustainable forest management (SFM). Yami Mastewal [12] considers the development of management plans and business model scenarios that balance the ecological and socio-economic goals at a local level in collaboration with rural communities is important to improve the governance of community-managed forests. Mann Carsten [13] combine several methodological approaches to understand the working conditions of such innovations, including integrated mapping, stakeholder and governance analysis, and a forward-looking scenario approach for innovation assessment. In terms of research methods, Foreign scholars mainly use interviews (Yuksel [14], 1999; Sheehan [15], 2005), questionnaire surveys (Sheehan [15], 2005), round table meetings (Ritchie [16], 2011), and other methods to collect the opinions of stakeholders to understand their attitudes and views on forestry development policies or measures. In the face of conflict and cooperation among stakeholders, the author often draws on the theories and methods of organization theory, communication methods, and public participation in sociology and management to analyze, so as to seek a theoretical framework to solve the problem and put forward countermeasures and suggestions for stakeholder management. For example, Markwick [17] (2000) believes that the power-interest matrix, a mature stakeholder analysis method, can be used to identify and explain the relationship between stakeholders.

In China, through the search of the China National Knowledge Network with the theme of “forest governance”, there are 460 literatures by 2023, mainly covering global forest governance and the theory of rights in forest governance. Long Hexing et al. [18] studied the mechanism and application of government control, market-based measures, community-based forest management, and mixed governance in the forest field and analyzed the driving force and performance of the rise of non-governmental forest governance subjects and forms in order to promote the construction of our forestry governance system. Zhao Jiacheng et al. [19] introduced the actor-centered power theory established by Max Krott et al., which proposed that actors could impose their will on other actors and change their behavior through the three power elements of coercion, incentive (inhibition), and dominant information, thus determining the final effect of forest governance. Zhu Zhenfeng et al. [20], based on the new situation and mission of forestry development during the 14th Five-Year Plan period, systematically reviewed the theoretical exploration and unremitting practice carried out by the Communist Party of China in the field of forest governance from the perspective of forest governance, taking the centennial of the founding of the Party as an opportunity. Huang Chong et al. [21] proposed the framework of social ecosystem analysis in the forest governance model and related studies, which provided a feasible path to explore the coupling effect of natural, political, economic, social, and other factors and helped to further explain various “failures” in the reality of many popular forest governance policies and management models. Su Bo [22] takes REDD from an analytical perspective and believes that Latin American countries have the great potential to eventually reduce deforestation and forest degradation and the corresponding capacity to mitigate climate change by utilizing their rich forest resources, but they need to firm up the determination of forest governance and reform and ensure the sustainability of REDD policy implementation.

Based on the above discussion, more comprehensive research on forest management has been carried out both at home and abroad. In forest governance, the government, communities, and enterprises consider interest demands, conflicts, and interactions, effectively build a tripartite evolutionary game model of the government, forestry enterprises, and rural community residents, analyze the evolution rules and stability strategies of game players, and motivate stakeholders to actively participate in forest governance.

3. Construction of a Dynamic Evolutionary Game Model of Forest Governance Core Stakeholder Relationship

Evolutionary game theory is based on traditional game theory and absorbs the ideas of biological evolution and gene theory to analyze economic phenomena with evolutionary ideas. Under the premise of incomplete information, bounded rational individuals/groups observe the decisions of other individuals, feel the external environment, constantly learn through trial and error, adjust and optimize their own decisions, and go through a series of dynamic processes. Finally, the optimal decision is made. In the process of evolutionary selection, based on game theory, because the opponent's strategy changes, it is necessary to adjust its own strategy to win. Evolutionary game theory pays attention to the dynamic adjustment process. Even if the optimal strategy has been determined, the adjustment of other individual strategies will still lead to a change in decision-making, and the cycle will advance, and finally achieve an evolutionary stable equilibrium.

Evolutionary game theory includes two core theories: evolutionarily stable strategy (ESS) and replicator dynamics (RD). Results: ESS can be defined as the need to meet two conditions: ① $E(S', S') > E(S', S)$; ② $E(S', S) > E(S, S)$ (E stands for returns, S' stands for strategies other than S). Results: Replicator Dynamics (RD) is a dynamic differential analysis method that is widely used to study the strategic stability of stakeholders, that is, a strategy that gains more returns than the average returns of other strategies during the game process and can evolve into a stable strategy through repeated games. Compared with traditional games, evolutionary games can explain how stakeholders achieve equilibrium in the long run, and stakeholders can adjust their own behavior by comparing the strategy choices of other game groups.

The core stakeholders in forest governance are the government, forestry enterprises, and rural community residents, who all play irreplaceable roles in the process of forest governance. At the same time, these three have their own roles, positioning, and interest demands, and there are conflicts of interest within and between groups. The sustainable and healthy development of forest resources requires the joint efforts, active participation, and effective cooperation of the three parties. It is necessary to constantly adjust each other's own strategies. The optimal strategy under the condition of bounded rationality can be found through evolutionary games to solve various conflicts of interest, which is conducive to the healthy development of forest resources. The optimal strategy selection of all parties needs to be analyzed by building a dynamic evolutionary game model.

Evolutionary game theory holds that the subject is bounded by rationality when making decisions. Due to information asymmetry and other reasons, the government, forestry enterprises, and rural community residents are unable to make the best response to the constantly changing reality. Therefore, they need to constantly adjust their own strategies and improve their own benefits through a series of measures such as observation, imitation, and trial and error in the decision-making process. Finally, we obtain equilibrium. A dynamic evolutionary game model of the tripartite relationship among government, forestry enterprises, and rural community residents is constructed, the evolutionary equilibrium strategy of each decision-making body is analyzed, and the stability of its results is discussed.

3.1. Basic Assumptions and Related Parameters

When choosing strategies, core stakeholders in forest governance will constantly adjust their own strategies according to the strategies of other stakeholders, which is a dynamic game process that is constantly optimized over time. In order to simplify the model, the following assumptions are made:

Hypothesis 1. In the natural environment, without considering other constraints, the government, enterprises, and rural community residents are regarded as a complete system, among which the three groups follow the basic assumption of bounded rationality and incomplete information symmetry. In the process of an evolutionary game, the three parties constantly adjust their behavior strategies to achieve the equilibrium of the “evolutionary game”.

Hypothesis 2. The policy set of the government is (guide, not guide), the policy set of the enterprise is (good faith, bad faith), and the policy set of the rural community residents is (participation, non-participation). Wherein, the probability that the government adopts a supervision, reward, and punishment guidance strategy is x . The probability of not guiding is $1 - x$. The probability that the enterprise adopts the honest management strategy is y , the probability of dishonest operations is $1 - y$, and the probability of rural community residents taking an active part in the forest management strategy is z . The probability of not participating is $1 - z$, where x , y , and z are functions t of time, and satisfy: $x, y, z \in [0, 1]$.

Hypothesis 3. The government actively guides forest governance, and the cost of supervision and inspection for enterprises is C_g . After forest management, remarkable ecological results have been achieved, which have been rewarded by the higher government V_g . If the government does not strictly implement supervision and management and is condemned by the public opinion, the credibility of the government will be lost, and the political cost is L_g .

Hypothesis 4. In the process of forest management, enterprises actively choose honest management, and the purchase of production equipment, maintenance, and personnel management requires input costs C_e . At this time, the production and operation income of the enterprise is R_e . When enterprises choose honest management, they are rewarded by the government V_e . If the enterprise chooses to operate in bad faith, it may obtain a higher additional income W_e . If the government adopts the guidance strategy of supervision and punishment or the rural community residents actively participate in the forest management strategy, the enterprise will be fined by the government P_e .

Hypothesis 5. The normal production income of rural community residents is R_f . Active participation in forest management, supervision, and reporting by enterprises needs to be paid C_f . It is possible that in the process of active participation in forest management, additional benefits can be obtained W_f . The enterprise may choose to operate in bad faith, resulting in the destruction of forest resources, such as woodland, and negative returns to the future production and lives of rural community residents D_f . The reward from the government for supervising and reporting enterprises is V_f . Residents of damaged rural communities received compensation from enterprises S_f .

3.2. Payment Function Construction

Based on the above assumptions and analysis, the payment matrix of the three-party game under different strategies can be given, as shown in Table 1.

Table 1. Tripartite game payment matrix.

		Government			
		Guide (x)		Unguided (1 – x)	
		Rural Community Residents			
		Participate in (z)	Non-Participation (1 – z)	Participate in (z)	Non-Participation (1 – z)
Enterprise	Good faith (y)	$R_e - C_e + V_e$	$R_e - C_e + V_e$	$R_e - C_e$	$R_e - C_e$
		$V_g - C_g - V_e$	$-C_g - V_e$	$V_g - L_g$	$-L_g$
	Dishonesty (1 – y)	$R_f + W_f - C_f$	R_f	$R_f + W_f - C_f$	R_f
		$R_e + W_e - P_e - S_f$	$R_e + W_e - P_e$	$R_e + W_e$	$R_e + W_e$
		$-C_g + P_e - V_f$	$-C_g + P_e$	$-L_g - V_f$	$-L_g$
		$R_f + W_f + S_f + V_f - C_f - D_f$	$R_f - D_f$	$R_f + W_f + S_f - C_f - D_f$	$R_f - D_f$

There are eight different strategy combinations in the game between the government, forestry enterprises, and rural community residents. In each strategy combination, each

side will obtain different benefits due to the different strategies of the other two sides. According to the above assumptions, the core stakeholders of each party are bounded by rationality, so each party will choose its own strategy according to the strategies of the other two parties.

4. Analysis of the Dynamic Evolutionary Game Model

4.1. Analysis of the Evolutionary Stability Strategy Based on a Replication Dynamic Equation

The interest relationship and strategy selection among the government, enterprises, and rural community residents are not invariable, and the strategies of each party will be adjusted with the changes in the strategies of the other two parties, which is a constantly evolving and gradually stable process. By replicating the analysis of the dynamic equation, a strategy with greater average returns than other strategies is determined. Evolutionary games form a stable strategy and can maintain a certain stability.

(1) Government evolution stability strategy

The expected earnings of the government choosing to guide and not guide are, respectively, U_{g1} , U_{g2} , the average expected return is U_g .

$$U_{g1} = yz(V_g - C_g - V_e) + y(1 - z)(-C_g - V_e) + z(1 - y)(-C_g + P_e - V_f) + (1 - y)(1 - z)(-C_g + P_e) \quad (1)$$

$$U_{g2} = yz(V_g - L_g) + y(1 - z)(-L_g) + z(1 - y)(-L_g - V_f) + (1 - z)(1 - y)(-L_g) \quad (2)$$

$$U_g = xU_{g1} + (1 - x)U_{g2} \quad (3)$$

Then, the government's replication dynamic equation is as follows:

$$F_{g(x)} = dx/dt = x(U_{g1} - U_g) = x(1 - x)(L_g + P_e - yP_e - yV_e - C_g) \quad (4)$$

The first derivative of Formula (4) is obtained:

$$F'_{g(x)} = (1 - 2x)(L_g + P_e - yP_e - yV_e - C_g) \quad (5)$$

According to the theory put forward by Friedman [23], if satisfied, $F_{g(x)} = 0$, $F'_{g(x)} < 0$, x represents the stability strategy (ESS) adopted by the government. The asymptotic stability analysis is conducted on the government's choice behavior strategy and evolution.

One is if $L_g + P_e - yP_e - yV_e - C_g = 0$, then $F_{g(x)} = 0$, represents the dividing line of stable states.

The second is if $L_g + P_e - yP_e - yV_e - C_g \neq 0$, make $F_{g(x)} = 0$, and obtain $x = 0$, $x = 1$, which are two stable points.

If $L_g + P_e - yP_e - yV_e - C_g < 0$, namely $y > \frac{L_g + P_e - C_g}{P_e + V_e}$, then there is $F'_{g(0)} < 0$, $F'_{g(1)} > 0$, so $x = 0$ is the stable strategy, which indicates that government guidance is in an unstable state and relaxed guidance is a stable state. Otherwise, if $L_g + P_e - yP_e - yV_e - C_g > 0$, namely $y < \frac{L_g + P_e - C_g}{P_e + V_e}$, then $F'_{g(0)} > 0$, $F'_{g(1)} < 0$, so $x = 1$ is the stable strategy, which indicates that government guidance is in a stable state and relaxed guidance is in an unstable state.

(2) Enterprise evolution stability strategy

The expected earnings of enterprises adopting honest and dishonest business strategies are U_{e1} , U_{e2} respectively, the average expected return is U_e .

$$U_{e1} = xz(R_e - C_e + V_e) + x(1 - z)(R_e - C_e + V_e) + z(1 - x)(R_e - C_e) + (1 - x)(1 - z)(R_e - C_e) \quad (6)$$

$$U_{e2} = xz(R_e + W_e - P_e - S_f) + x(1 - z)(R_e + W_e - P_e) + z(1 - x)(R_e + W_e) + (1 - x)(1 - z)(R_e + W_e) \quad (7)$$

$$U_e = yU_{e1} + (1 - y)U_{e2} \quad (8)$$

Then, the replication dynamic equation of the enterprise is as follows:

$$F_{e(y)} = dy/dt = y(U_{e1} - U_e) = y(1-y)(xzS_f + xV_e + xP_e - C_e - W_e) \quad (9)$$

The first derivative of Formula (9) can be obtained:

$$F'_{e(y)} = (1-2y)(xzS_f + xV_e + xP_e - C_e - W_e) \quad (10)$$

If satisfied $F_{e(y)} = 0$, $F'_{e(y)} < 0$, then y represents the stability strategy (ESS) adopted by the enterprise. This paper analyzes the asymptotic stability of the enterprise choice behavior strategy and its evolution.

If $xzS_f + xV_e + xP_e - C_e - W_e = 0$, namely $z = \frac{W_e + C_e - xP_e - xV_e}{xS_f}$, then $F_{e(y)} = 0$, denotes the stable state boundary.

If $xzS_f + xV_e + xP_e - C_e - W_e \neq 0$, make $F_{e(y)} = 0$, and obtain $y = 0$, $y = 1$, which are two stable points.

If $xzS_f + xV_e + xP_e - C_e - W_e < 0$, namely $z < \frac{W_e + C_e - xP_e - xV_e}{xS_f}$, then $F'_{e(0)} < 0$, $F'_{e(1)} > 0$, so $y = 0$ is the stable strategy, which indicates that the honest management is in an unstable state, and the dishonest management is in a stable state. Conversely, if $xzS_f + xV_e + xP_e - C_e - W_e > 0$, namely $z > \frac{W_e + C_e - xP_e - xV_e}{xS_f}$, Then there is $F'_{e(0)} > 0$, $F'_{e(1)} < 0$, so $y = 1$ is the stable strategy, it indicates that honest operation is in a stable state, and dishonest operation is in an unstable state.

(3) Rural community residents evolutionary stability strategy

The expected benefits of rural community residents participating in forest management and not participating in forest management are U_{f1} and U_{f2} , respectively, The average expected return is U_f , as follows:

$$U_{f1} = xy(R_f + W_f - C_f) + y(1-x)(R_f + W_f - C_f) + x(1-y)(R_f + W_f + S_f + V_f - C_f - D_f) + (1-x)(1-y)(R_f + W_f + S_f - C_f - D_f) \quad (11)$$

$$U_{f2} = xyR_f + y(1-x)R_f + x(1-y)(R_f - D_f) + (1-x)(1-y)(R_f - D_f) \quad (12)$$

$$U_f = zU_{f1} + (1-z)U_{f2} \quad (13)$$

Then, the replication dynamic equation of rural community residents is as follows:

$$F_{f(z)} = dz/dt = z(U_{f1} - U_f) = z(1-z)(xV_f - xyV_f + W_f + S_f - C_f - yS_f) \quad (14)$$

The first derivative of Formula (14) can be obtained:

$$F'_{f(z)} = (1-2z)(xV_f - xyV_f + W_f + S_f - C_f - yS_f) \quad (15)$$

If satisfied, $F_{f(z)} = 0$, $F'_{f(z)} < 0$, then z represents the stability strategies (ESS) adopted by residents in rural communities and analyzes the gradual stability of the choice behavior strategy and evolution of rural community residents.

One is if $xV_f - xyV_f + W_f + S_f - C_f - yS_f = 0$, then $F_{f(z)} = 0$, denotes the boundary between steady states.

The second is if $xV_f - xyV_f + W_f + S_f - C_f - yS_f \neq 0$, make $F_{f(z)} = 0$, and obtain $z = 0$, $z = 1$, which are two stable points.

If $xV_f - xyV_f + W_f + S_f - C_f - yS_f < 0$, namely $x < \frac{yS_f + C_f - W_f - S_f}{V_f(1-y)}$, then $F'_{f(0)} < 0$, $F'_{f(1)} > 0$, so $z = 0$ is the stable strategy, which indicates that the participation of rural community residents in governance is in an unstable state, while the non-participation in governance is in a stable state. Conversely, if $xV_f - xyV_f + W_f + S_f - C_f - yS_f > 0$, namely $x > \frac{yS_f + C_f - W_f - S_f}{V_f(1-y)}$, then there is $F'_{f(0)} > 0$, $F'_{f(1)} < 0$, so $z = 1$ is the stable strategy,

which indicates that the participation of rural community residents in governance is in a stable state, while the non-participation of rural community residents in governance is in an unstable state.

Through the analysis of the above three subjects' choice behavior strategies and the evolutionary stability of the government, enterprises, and rural community residents, it can be concluded that:

First, the evolutionarily stable state of government policy choice is affected by the change in the proportion of honest management.

Second, the evolutionarily stable state of firm strategy choice is influenced by the proportion of government guidance and the proportion of rural community residents choosing to participate in governance.

Thirdly, the evolutionary stability of the strategy choice of rural community residents is affected by the proportion of government guidance and the proportion of honest management of enterprises.

Through adjustment, the game between the government, enterprises, and rural community residents will eventually evolve into a stable state of active guidance, honest management, and participation in governance. The following points should be achieved:

First, when $y < \frac{L_g + P_e - C_g}{P_e + V_e}$, $x \rightarrow 1$, namely, the government actively guided and adopted strict regulatory strategies. Therefore, reducing the cost C_g of government supervision, increasing the government's fine P_e for disintegrity enterprises, increasing the government's political cost L_g , and improving the government's guidance V_e for integrity enterprises can prompt the government to choose the guiding strategy of strict supervision.

Second, when $z > \frac{W_e + C_e - xP_e - xV_e}{xS_f}$, $y \rightarrow 1$, namely, the enterprise chooses the integrity management strategy. Therefore, by reducing the input cost C_e of enterprises, increasing the fines P_e imposed by the government on disintegrity businesses, improving the government's guidance V_e for integrity enterprises, increasing the compensation S_f for damaged farmers, increasing the probability x of strict supervision by the government on enterprises, and reducing the extra income W_e of enterprises operating in bad faith, enterprises will finally choose to operate in good faith.

Third, when $x > \frac{yS_f + C_f - W_f - S_f}{V_f(1-y)}$, $z \rightarrow 1$, namely, rural community residents take an active part in the governance strategy. Therefore, by enhancing the enthusiasm of rural community residents to participate in forest management the pay costs due to supervision whistleblowing enterprises is expressed in terms of C_f increasing the additional income W_f that may be obtained in the process of active participation in forest management, increasing the compensation S_f for damaged farmers, and improving the supervision and reporting of dishonest enterprises to obtain rewards V_f from local governments, all of which help rural community residents choose to actively participate in forest management.

4.2. Stability Analysis of the Equilibrium Point

Based on Equations (4), (9), and (14), the three-dimensional replication dynamic equation system of government, enterprises, and rural community residents is obtained:

$$\begin{cases} F_g(x) = dx/dt = x(U_{g1} - U_g) = x(1-x)(L_g + P_e - yP_e - yV_e - C_g) \\ F_e(y) = dy/dt = y(U_{e1} - U_e) = y(1-y)(xzS_f + xV_e + xP_e - C_e - W_e) \\ F_f(z) = dz/dt = z(U_{f1} - U_f) = z(1-z)(xV_f - xyV_f + W_f + S_f - C_f - yS_f) \end{cases}$$

The above dynamic three-dimensional system describes the dynamic adjustment process of strict supervision, honest management, and participation strategy selection by the government, enterprises, and rural community residents until Nash equilibrium is reached.

Making $F_g(x) = 0$, $F_e(y) = 0$, $F_f(z) = 0$, eight pure strategy local equilibrium points can be obtained: $E_1(0, 0, 0)$, $E_2(0, 0, 1)$, $E_3(0, 1, 0)$, $E_4(1, 0, 0)$, $E_5(1, 1, 0)$, $E_6(1, 0, 1)$, $E_7(0, 1, 1)$,

and $E_8(1,1,1)$. There is also a mixed strategy local equilibrium point $E_9(x^*, y^*, z^*)$. Equation (16) is satisfied, and the range is $0 < x^* < 1, 0 < y^* < 1, 0 < z^* < 1$.

$$\begin{cases} L_g + P_e - yP_e - yV_e - C_g = 0 \\ xzS_f + xV_e + xP_e - C_e - W_e = 0 \\ xV_f - xyV_f + W_f + S_f - C_f - yS_f = 0 \end{cases} \quad (16)$$

It can be obtained from the above equation:

$$\begin{cases} x^* = \frac{(W_f - C_f)(P_e + V_e)}{(L_g - C_e - V_e)V_f} - \frac{S_f}{V_f} \\ y^* = \frac{L_g + P_e - C_e}{P_e + V_e} \\ z^* = \frac{(W_e + C_e)(L_g - C_e - V_e)V_f}{S_f(W_f - C_f)(P_e + V_e) - S_f^2(L_g - C_e - V_e)} - \frac{P_e + V_e}{S_f} \end{cases}$$

Friedman believes that the stability of the population dynamic equilibrium described by the differential equation system can be obtained from the local stability analysis of the Jacobian matrix of the system. According to Lyaplov's first rule, the Jacobian matrix of a three-dimensional replication dynamic equation system can be obtained by differentiating each dynamic replication equation:

$$J = \begin{bmatrix} \frac{\partial F_g(x)}{\partial x} & \frac{\partial F_g(x)}{\partial y} & \frac{\partial F_g(x)}{\partial z} \\ \frac{\partial F_e(y)}{\partial x} & \frac{\partial F_e(y)}{\partial y} & \frac{\partial F_e(y)}{\partial z} \\ \frac{\partial F_f(z)}{\partial x} & \frac{\partial F_f(z)}{\partial y} & \frac{\partial F_f(z)}{\partial z} \end{bmatrix} = \begin{bmatrix} (1-2x)(L_g + P_e) & x(1-x)(-P_e - V_e) & 0 \\ -yP_e - yV_e - C_g & (1-2y)(xzS_f + xV_e + xP_e - C_e - W_e) & y(1-y)xS_f \\ y(1-y)(zS_f + V_e + P_e) & z(1-z)(-xV_f - S_f) & (1-2z)(xV_f - xyV_f + W_f + S_f - C_f - yS_f) \end{bmatrix}$$

Take $E_1(0,0,0)$ as an example to discuss its asymptotic stability condition, from J

we obtain the Jacobian $J = \begin{bmatrix} L_g + P_e - C_g & 0 & 0 \\ 0 & -C_e - W_e & 0 \\ 0 & 0 & W_f + S_f - C_f \end{bmatrix}$ at the equilibrium

point $E_1(0,0,0)$. In this case, the eigenvalue of matrix J is $\lambda_1 = L_g + P_e - C_g$, $\lambda_2 = -C_e - W_e$, $\lambda_3 = W_f + S_f - C_f$. If satisfied $\lambda_1 < 0$, $\lambda_2 < 0$, $\lambda_3 < 0$, namely $L_g + P_e - C_g < 0$, $-C_e - W_e < 0$, $W_f + S_f - C_f < 0$, then $E_1(0,0,0)$ is an evolutionarily stable equilibrium solution.

The Jacobian at the equilibrium point $E_3(0,1,0)$ is

$J = \begin{bmatrix} L_g - V_e - C_g & 0 & 0 \\ 0 & -(-C_e - W_e) & 0 \\ 0 & 0 & W_f - C_f \end{bmatrix}$. This is the eigenvalue $\lambda_2 > 0$ in J. The

stability condition is not satisfied, so $E_3(0,1,0)$ is an unstable point. In the same way, $E_7(0,1,1)$ and $E_9(x^*, y^*, z^*)$ are also unstable points. Table 2 lists the stability conditions of other equilibrium points.

Table 2. Equilibrium stability conditions of three-dimensional replica dynamic equation systems.

Equilibrium Point	Stability Condition
$E_1(0,0,0)$	$L_g + P_e - C_g < 0, -C_e - W_e < 0, W_f + S_f - C_f < 0$
$E_2(0,0,1)$	$L_g + P_e - C_g < 0, -C_e - W_e < 0, -(W_f + S_f - C_f) < 0$
$E_4(1,0,0)$	$-(L_g + P_e - C_g) < 0, V_e + P_e - C_e - W_e < 0, V_f + W_f + S_f - C_f < 0$
$E_5(1,1,0)$	$-(L_g - V_e - C_g) < 0, -(V_e + P_e - C_e - W_e) < 0, W_f - C_f < 0$
$E_6(1,0,1)$	$-(L_g + P_e - C_g) < 0, S_f + V_e + P_e - C_e - W_e < 0, -(V_f + W_f + S_f - C_f) < 0$
$E_8(1,1,1)$	$-(L_g - V_e - C_g) < 0, -(S_f + V_e + P_e - C_e - W_e) < 0, -(W_f - C_f) < 0$

If the government, enterprises, and rural community residents participate in forest management, so $E_8(1,1,1)$ is ESS. The stability conditions in Table 2 show that when the political cost of the government is greater than the guidance cost of the government and the reward to the enterprises operating in good faith, the government's fine to the enterprises operating in bad faith, the reward to the enterprises operating in good faith, and the compensation to the residents of the damaged rural communities are greater than the additional income of the enterprises operating in bad faith and the cost of the enterprises participating in governance. In addition, when the additional benefits of rural community residents' participation in governance are greater than the costs of participating, supervising, reporting, etc., the three parties will be profitable in forest governance. At this time, the three parties will cooperate and finally choose the behavioral strategy (guidance, integrity, and participation) to enter a virtuous cycle of forest governance.

5. Numerical Simulation

The stability of the dynamic evolution system of the government, forestry enterprises, and rural community residents can be obtained from the analysis of equilibrium points. In order to illustrate the stability of the strategy evolution of the government, forestry enterprises, and rural community residents more intuitively, the dynamic evolution process of their strategies is simulated by parameter assignment.

Parameter assignment needs to be considered in combination with the actual situation. For example, when forestry enterprises choose to operate in good faith, the cost of purchasing production equipment, maintenance, and personnel management is higher, and the fair sale of forest products, rather than a malicious price increase, will obtain less benefits than a dishonest operation, so assume $R_e < W_e$. At the same time, considering the government's guidance, the participation of rural community residents, and the integrity of the management of forestry enterprises, three sets of values were taken for simulation. Each group was assigned a value between 1 and 10, considering various actual conditions. Matlab 2016a software is used to simulate the dynamic evolution process of the tripartite strategy.

5.1. The Analysis of the Evolution Stability of the Government Guidance Strategy Is Highlighted

Considering the strict supervision and punishment after government guidance, the P_e value is 7, and the parameter assignment is as follows: $C_g = 4$, $V_g = 5$, $L_g = 6$, $C_e = 5$, $R_e = 6$, $V_e = 1$, $W_e = 7$, $P_e = 7$, $C_f = 4$, $W_f = 6$, $D_f = 3$, $V_f = 4$, $S_f = 3$, $E_1(0,0,0)$, $E_2(0,0,1)$, $E_4(1,0,0)$, $E_5(1,1,0)$, $E_7(0,1,1)$, $E_8(1,1,1)$ are calculated as the saddle point, $E_3(0,1,0)$ is the unstable point, $E_6(1,0,1)$ is ESS point, Table 3 lists the results. It is calculated that $E_9(-2.083, 0.5, -5.227)$ has no particularity, so it is not listed in the table.

Table 3. Highlights the analysis of the stability results of government guidance strategies.

Equilibrium Point	Result	Eigenvalue
$E_1(0,0,0)$	Saddle point	9, −12, 5
$E_2(0,0,1)$	Saddle point	9, −12, −5
$E_3(0,1,0)$	Unstable point	1, 12, 2
$E_4(1,0,0)$	Saddle point	−9, −4, 9
$E_5(1,1,0)$	Saddle point	−1, 4, 2
$E_6(1,0,1)$	ESS	−9, −1, −9
$E_7(0,1,1)$	Saddle point	1, 12, −2
$E_8(1,1,1)$	Saddle point	−9, 1, −2
$E_9(-2.083, 0.5, -5.227)$	—	—

As can be seen from Table 3, under the guidance of the government, heavy punishment should be given to forestry enterprises that operate in bad faith and active participation of rural community residents. (1,0,1), that is, (guidance, bad faith, and participation) is the final stable strategy. After the three parties go through the dynamic game process,

there will be a stable state in which the government guides forest governance, forestry enterprises operate in bad faith, and rural community residents participate.

5.2. Highlight the Evolution and Stability Analysis of Corporate Integrity Strategy

Considering that the integrity management of forestry enterprises can obtain better social response, support from rural community residents, increase sales, and benefit from integrity, the V_e value is 3, and the parameter assignment is as follows: $C_g = 2$, $V_g = 5$, $L_g = 6$, $C_e = 2$, $R_e = 4$, $V_e = 3$, $W_e = 8$, $P_e = 3$, $C_f = 4$, $W_f = 3$, $D_f = 2$, $V_f = 2$, $S_f = 3$, $E_1(0, 0, 0)$, $E_2(0, 0, 1)$, $E_3(0, 1, 0)$, $E_4(1, 0, 0)$, $E_5(1, 1, 0)$, $E_8(1, 1, 1)$ are calculated as the saddle point, $E_7(0, 1, 1)$ is the unstable point, $E_6(1, 0, 1)$ is ESS point. Table 4 lists the results. It is calculated that $E_9(-4.5, 1.167, -3.74)$ has no particularity, so it is not listed in the table.

Table 4. Analysis of the stability results of the business integrity strategy.

Equilibrium Point	Result	Eigenvalue
$E_1(0, 0, 0)$	Saddle point	7, −10, 2
$E_2(0, 0, 1)$	Saddle point	7, −10, −2
$E_3(0, 1, 0)$	Saddle point	1, 9, −1
$E_4(1, 0, 0)$	Saddle point	−7, −4, 4
$E_5(1, 1, 0)$	Saddle point	−1, 4, −1
$E_6(1, 0, 1)$	ESS	−7, −4, −4
$E_7(0, 1, 1)$	Unstable point	1, 10, 1
$E_8(1, 1, 1)$	Saddle point	−1, 1, 1
$E_9(-4.5, 1.167, -3.74)$	—	—

As can be seen from Table 4, when forestry enterprises operate in bad faith under government guidance, rural community residents actively participate, supervise and report the dishonest behaviors of forestry enterprises, and supervise the behaviors that the government does not guide. In addition, (1,0,1) (guidance, dishonesty, and participation) is a stable strategy. After a dynamic game among the three parties, the government will tend not to guide forest management, forestry enterprises will not operate in good faith, and rural community residents will participate in forest management.

5.3. Analysis on the Evolution Stability of Rural Community Residents' Participation Strategy

Considering that rural community residents actively participate in forest management, supervise and report dishonest business practices of enterprises, and receive greater rewards, the V_f value is 8, and the parameter assignment is as follows: $C_g = 3$, $V_g = 5$, $L_g = 6$, $C_e = 5$, $R_e = 6$, $V_e = 2$, $W_e = 7$, $P_e = 8$, $C_f = 3$, $W_f = 7$, $D_f = 3$, $V_f = 8$, $S_f = 3$, $E_1(0, 0, 0)$, $E_2(0, 0, 1)$, $E_4(1, 0, 0)$, $E_5(1, 1, 0)$, $E_6(1, 0, 1)$, $E_7(0, 1, 1)$ are calculated as the saddle point, $E_3(0, 1, 0)$ is the unstable point, $E_8(1, 1, 1)$ is ESS point. Table 5 lists the results. It is calculated that $E_9(-5.375, 0.900, -4.077)$ has no particularity, so it is not listed in the table.

Table 5. Analysis of the stability results of rural community residents' participation strategies.

Equilibrium Point	Result	Eigenvalue
$E_1(0, 0, 0)$	Saddle point	11, −12, 7
$E_2(0, 0, 1)$	Saddle point	11, −12, −7
$E_3(0, 1, 0)$	Unstable point	1, 12, 4
$E_4(1, 0, 0)$	Saddle point	−11, −2, 15
$E_5(1, 1, 0)$	Saddle point	−1, 2, 4
$E_6(1, 0, 1)$	Saddle point	−11, 1, −15
$E_7(0, 1, 1)$	Saddle point	1, 12, −4
$E_8(1, 1, 1)$	ESS	−1, −1, −4
$E_9(-5.375, 0.900, -4.077)$	—	—

As can be seen from Table 5, in the case of government guidance, rural community residents actively participate in the phenomenon of honest management of forestry enterprises, supervise and report dishonest behaviors of forestry enterprises, and supervise the behaviors that the government does not guide. In addition, (1,1,1) namely (guidance, integrity, and participation) is a stable strategy. After the dynamic game between the three parties, the government will guide forest management, forestry enterprises will operate with integrity, and rural community residents will participate in forest management.

In the stability analysis of the three numerical simulations, there are unstable points (0,1,0) and (0,1,1); that is, the strategy combination of government not guiding, forestry enterprises operating with integrity, and rural community residents participating or not participating in forest governance is unstable. It proves that the non-guidance strategy adopted by the government cannot achieve the balance of interests of the core stakeholders in forest governance. In the three kinds of steady-state strategy sets of numerical simulation, the government takes the guiding strategy. The government plays a prominent and indispensable role in guiding and rewarding honest forestry enterprises and rural community residents who participate in forest management.

6. Conclusions and Suggestion

In this paper, a tripartite evolutionary game model of government, forestry enterprises, and rural community residents is constructed, and the stability strategy of each interest subject is analyzed and simulated. The research shows that the core stakeholders of forest governance, the government, forestry enterprises and rural community residents can choose from the strategies of “guiding/not guiding”, “good faith/bad faith” and “participating/not participating”, forming a total of eight strategy combinations. All three are bounded by rationality and limited by cognition, so strategy selection is a dynamic adjustment process. At the same time, the strategic choices of the other two parties are affected. The “guidance” of the government is a strong guarantee for the realization of triateral equilibrium, and the government is in a very active and strong position (Lu Xiaoli et al. [24]). If the government chooses not to guide the strategy, the instability points (0,1,0) and (0,1,1) will appear in the three numerical simulation situations, which require the guidance of the government to achieve stability. The government’s choice of “guidance” strategy has no significant impact on whether forestry enterprises choose the “honesty” strategy, but it has a significant impact on rural community residents’ participation in strategy selection. The government encourages residents of rural communities to take an active part in forest management and supervises whether forestry enterprises operate with integrity. After the government chooses the “guidance” strategy, it promotes the honest management of forestry enterprises, increases the reward for the participation of rural community residents in forest management, and finally achieves the balance of interests for the three parties. Forestry enterprises are the practitioners of forest governance, but they lack dominant power and are in a weak position. They can only exert influence to a certain extent but cannot change the trend of evolution. The “guidance” of the government and the active participation of rural community residents can finally achieve a stable state. Accordingly, this paper puts forward suggestions on forest management from the perspectives of the government, forestry enterprises, and rural community residents:

(1) The government should attach great importance to the management of forest resources and encourage forest-related enterprises and the public to protect and manage forest resources. At the same time, increase the punishment for the destruction of forest resources, improve the supervision system, and reward policies.

(2) Forestry enterprises should actively respond to the call of the national forest resources protection and management policy, actively introduce forestry professionals and technical personnel, innovate the way of forest resources utilization, change the traditional extensive production mode, and improve the level of enterprise operation and management.

(3) Rural community residents should actively participate in the protection of forest resources, strengthen the supervision and reporting of the surrounding forestry enterprises

by means of online platforms, improve their own awareness of forest resource protection, and actively participate in forest management.

This paper proposes a dynamic evolutionary game model to analyze the dynamic process of the tripartite game between the government, forestry enterprises, and rural community residents. Due to the limitations of its own level, only the three-party dynamic game process is analyzed. Although it has improved compared with the two-player game in previous studies, other interested players are ignored, weakening the scientific results of the game. In the next step, dynamic parameters of randomness can be introduced to make the results of the evolutionary game more scientific.

Funding: Project was supported by the Scientific Research Foundation of the Higher Education Institutions of Hebei Province, China (Grant No. BJK2023045).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

References

- Hengst-Ehrhart, Y.; Schraml, U. Back to the Forest's future: Guiding principles of German forest stakeholders and their impact on the forestry sector. *Land Use Policy* **2020**, *94*, 10449. [\[CrossRef\]](#)
- Salam, M.A.; Noguchi, T. Evaluating capacity development for participatory forest management in Bangladesh's Sal forests based on '4Rs' stakeholder analysis. *For. Policy Econ.* **2006**, *8*, 785–796. [\[CrossRef\]](#)
- Schusser, C.; Krott, M.; Movuh, M.C.Y.; Logmani, J.; Devkota, R.R.; Maryudi, A.; Salla, M.; Bach, N.D. Powerful stakeholders as drivers of community forestry—Results of an international study. *For. Policy Econ.* **2015**, *58*, 92–101. [\[CrossRef\]](#)
- Kumar, S.; Kant, S. Exploded logit modeling of stakeholders' preferences for multiple forest values. *For. Policy Econ.* **2007**, *9*, 516–526. [\[CrossRef\]](#)
- Sarvestani, A.A.; Ingram, V. Perceptions and practices of rural council participatory forest governance: Closed co-management in Chehel-Chay, Iran. *For. Policy Econ.* **2020**, *117*, 102202. [\[CrossRef\]](#)
- Arts, B. Assessing forest governance from a 'Triple G' perspective: Government, governance, governmentality. *For. Policy Econ.* **2014**, *49*, 17–22. [\[CrossRef\]](#)
- Juerges, N.; Arts, B.; Masiero, M.; Başkent, E.Z.; Borges, J.G.; Brodrechtova, Y.; Brukas, V.; Canadas, M.J.; Carvalho, P.O.; Corradini, G.; et al. Integrating ecosystem services in power analysis in forest governance: A comparison across nine European countries. Science Direct. *For. Policy Econ.* **2020**, *121*, 102317. [\[CrossRef\]](#)
- Agbo, V.M.; Swatuk, L. From conflict to collaboration: Atewa Forest governance. In *Natural Resources Forum*; Blackwell Publishing Ltd.: Oxford, UK, 2023; Volume 47, pp. 134–151.
- Basil Ewane, E. Forest Governance Effectiveness of Community- and Government-managed Forests in Cameroon. *Int. J. Environ. Clim. Change* **2022**, *12*, 83–102. [\[CrossRef\]](#)
- Salo, M.; Hiedanpää, J.; Orihuela, J.C.; Pinto, C.A.L.; Vetter, J.L. Governmentality in evidence? Evolving rationalities of forest governance in Peru. *Land Use Policy* **2023**, *129*, 106622. [\[CrossRef\]](#)
- Cadman, T.; Maraseni, T.; Koju, U.A.; Shrestha, A.; Karki, S. Forest Governance in Nepal concerning Sustainable Community Forest Management and Red Panda Conservation. *Land* **2023**, *12*, 493. [\[CrossRef\]](#)
- Yami, M.; Mekuria, W. Challenges in the Governance of Community-Managed Forests in Ethiopia: Review. *Sustainability* **2022**, *14*, 1478. [\[CrossRef\]](#)
- Mann, C.; Loft, L.; Hernández-Morcillo, M. Assessing forest governance innovations in Europe: Needs, challenges and ways forward for sustainable forest ecosystem service provision. *Ecosyst. Serv.* **2021**, *52*, 101384. [\[CrossRef\]](#)
- Yuksel, A.; Yuksel, F.; Bramwell, B. Stakeholder interviews and tourism planning at Pamukkale, Turkey. *Tour. Manag.* **1999**, *20*, 351–360. [\[CrossRef\]](#)
- Sheehan, L.R.; Ritchie, J. Destination Stakeholders Exploring Identity and Salience. *Ann. Tour. Res.* **2005**, *32*, 711–734. [\[CrossRef\]](#)
- Gomezelj Omerzel, D. Stakeholders' Understanding of Factors Influencing Tourism Demand Conditions: The case of Slovenia. *Tour. Hosp. Manag.* **2011**, *17*, 78–81. [\[CrossRef\]](#)
- Markwick, M.C. Golf tourism development, stakeholders, differing discourses and alternative agendas: The case of Malta. *Tour. Manag.* **2000**, *21*, 515–524. [\[CrossRef\]](#)
- Long, H.; Zhang, M.; Liu, J. From regulation to governance: The rise of forest governance. *For. Econ.* **2016**, *38*, 19–24.
- Zhao, J.; Zhang, P.; Ba, F.; Max, K.; Liu, J. Actor-centered power theory in forest governance. *Resour. Sci.* **2020**, *42*, 636–648.
- Zhu, Z.; Guan, J.; Feng, H.; Chen, L. The historical change and development practice of the Communist Party of China's forest management concept. *Issues For. Econ.* **2021**, *41*, 337–343.

21. Huang, C.; Luo, P.Z. Application and enlightenment of social ecosystem analysis framework in global forest governance research. *World For. Res.* **2022**, *35*, 1–7.
22. Su, B. Three models of Latin American countries' participation in global forest governance: A case study of REDD. *World For. Res.* **2023**, *4*, 117–122.
23. Friedman, D. Evolutionary Games in Economics. *Econometrica* **1991**, *59*, 637–666. [[CrossRef](#)]
24. Lu, X.; Mao, Y.; Gan, J. Analysis on the interest rank and balance of rural tourism stakeholders. *Resour. Dev. Mark.* **2017**, *9*, 1134–1137.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.